

Kento Tomita

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RESEARCH INTERESTS

My research interests lie primarily in uncertainty quantification and solving planning problems under uncertainty, enabling robots or operators to make safe decisions with sufficient confidence. My research leverages tools and techniques from optimization, optimal control, machine learning, deep neural networks, statistics, and dynamical systems.

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Ph.D. in Aerospace Engineering — AUG 2019 – *Present*

Cumulative GPA: 3.8 / 4.0, Advisor: Dr. Koki Ho

Topic: Hazard Detection and Avoidance for Autonomous Safe Planetary Landing

The University of Tokyo, Tokyo, Japan

Enrolled in M.E. Program in Aerospace Engineering — APR 2018 – MAR 2019

Cumulative GPA: 4.0 / 4.0, Advisor: Dr. Ryu Funase

ADCS Engineering for 6U Deep Space Cubesat, **EQUULEUS**

Kyushu University, Fukuoka, Japan

B.S. in Aerospace Engineering — APR 2014 – MAR 2018

Cumulative GPA: 3.8 / 4.0, Advisor: Dr. Mai Bando, Research Field: Astrodynamics

Thesis: Network Analysis of Space Debris in Low Earth Orbits

RESEARCH EXPERIENCE

Graduate Research Assistant

Present
AUG 2019

Space Systems Optimization Group, GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA

- Sensor Tasking for Space Domain Awareness
 - Proposed a novel and efficient mixed integer linear programming formulation for optimal sensor tasking in cislunar space via extended information filter.
 - Analyzed and related the Fisher information of optical measurements with the uncertainty deformation caused by cislunar dynamics
 - Developed space-borne sensor tasking simulator in Python
- Perception-Aware Optimal Guidance for Safe Planetary Landing
 - Enabled perception-aware optimal guidance via a surrogate model of reachable set/visible space based on deep neural network.
 - Developed efficient and theory-proved data generation pipeline for reachable set via lossless convexification and convex optimization for planetary landing dynamics.
 - Formulated perception-aware optimal guidance problem for safe planetary landing and solved with deep reinforcement learning.
- Hazard Detection Algorithm for Safe Planetary Landing
 - Developed a novel uncertainty quantification algorithm for landing safety probability under topographic uncertainty via Gaussian random field/2D Gaussian process model.
 - Developed and demonstrated uncertainty-aware hazard detection algorithm for LiDAR point cloud data and optical imagery with deep semantic segmentation.
 - Developed terrain and safety map data generation pipeline with high performance computing.
- Research Initiatives
 - Helped writing of the selected proposal for **NASA Early Career Faculty Award, 2019**.
 - Planned, completed, and documented 4 years of research project as a student lead.
 - Mentored total of 4 graduate and 4 undergraduate students, resulting in 1 journal, 2 proceedings, and 1 Master thesis with them being first authors.

Graduate Research Assistant

MAR 2019 | [Intelligent Space Systems Laboratory](#), Tokyo, Japan

- APR 2018
- Worked as Attitude, Dynamics, and Control System Engineer of [EQUULEUS](#) (Launched with NASA Artemis-1 Mission)
 - Designed, integrated, and tested ADCS module
 - Conducted Fault Tree Analysis of the Delta-V operations for lunar flyby

PUBLICATIONS

Journal Papers

- [J1] **K. Tomita**, K. Skinner, and K. Ho, "[Bayesian Deep Learning for Segmentation for Autonomous Safe Planetary Landing](#)," *Journal of Spacecraft and Rockets*, Vol. 59, No. 6, pp. 1800-1808, 2022.
- [J2] T. Claudet, **K. Tomita**, and K. Ho, "[Benchmark Analysis of Semantic Segmentation Algorithms for Safe Planetary Landing Site Selection](#)," *IEEE Access*, Vol. 10, No. 6, pp. 41766-41775, 2022.
- [J3] K. Ho, R. Beeson, **K. Tomita**, O. Gunasekara, and A. J. Sinclair, "[Analysis of Information-Theoretic Initial Sensor Search Method for Space Situation Awareness](#)," *Journal of Guidance, Control, and Dynamics*, Vol. 44, No. 3, pp. 641-645, 2021.
- [J4] R. Funase, **K. Tomita**, et.al., "[Mission to Earth-Moon Lagrange Point by a 6U CubeSat: EQUULEUS](#)," *IEEE Aerospace and Electronic Systems Magazine*, Vol. 35, No. 3, pp. 30-44, 2020.

Conference Papers

- [C1] **K. Tomita**, Y. Shimane, and K. Ho, "[Multi-Spacecraft Predictive Sensor Tasking for Cislunar Space Situational Awareness](#)," *AMOS Conference*, Maui, Hawaii, September 2023.
- [C2] Y. Shimane, **K. Tomita**, and K. Ho, "Strategic Regions for Monitoring Incoming Low-Energy Transfers to Low-Lunar Orbits," *AMOS Conference*, Maui, Hawaii, September 2023.
- [C3] **K. Tomita** and K. Ho, "[Learning Reachability for Hazard Detection and Avoidance in Planetary Landing](#)," *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, MT, August 2023.
- [C4] I. Nulton, **K. Tomita**, Y. Shimane, and K. Ho, "Sensitivity Analysis of Separation Time along Weak Stability Boundary Transfers," *AAS/AIAA Astrodynamics Specialist Conference*, Big Sky, MT, August 2023.
- [C5] **K. Tomita** and K. Ho, "[Stochastic Hazard Detection for Landing Under Topographic Uncertainty](#)," *AAS/AIAA Space Flight Mechanics Meeting*, Austin, TX, January 2023.
- [C6] **K. Tomita***, T. Driver*, and K. Ho, "[Deep Monocular Hazard Detection for Safe Small Body Landing](#)," *AAS/AIAA Space Flight Mechanics Meeting*, Austin, TX, January 2023. * These authors contributed equally to this work.
- [C7] **K. Tomita**, Y. Shimane, and K. Ho, "Small body reconnaissance by multiple spacecraft via deep reinforcement learning," *AAS/AIAA Astrodynamics Specialist Conference*, Charlotte, NC, August 2022.
- [C8] **K. Tomita** and K. Ho, "Adaptive Hazard Detection and Avoidance for Planetary Landing via Bayesian Semantic Segmentation," *AIAA SciTech 2022 Forum*, San Diego, CA, January 2022.
- [C9] **K. Tomita**, K. A. Skinner, K. Iiyama, B. A. Jagatia, T. Nakagawa, and K. Ho, "Hazard Detection Algorithm for Planetary Landing Using Semantic Segmentation," *ASCEND*, Las Vegas, NV, November 2020.
- [C10] K. Iiyama, **K. Tomita**, B. A. Jagatia, T. Nakagawa, and K. Ho, "Deep Reinforcement Learning for Safe Landing Site Selection with Concurrent Consideration of Divert Maneuvers," *AAS/AIAA Astrodynamics Specialist Conference*, Lake Tahoe, CA, August 2020.

SKILLS

Programming Languages: Python, C++/C, MATLAB, Fortran
Software: Pytorch, Gurobi, scikit-learn, Gym, SPICE, Trimesh, Blender, OpenCV, CUDA
Languages: ENGLISH (business), JAPANESE (native)

AWARDS AND SCHOLARSHIPS

NOV 2022 | Excellent Reviewer for the Journal of Guidance, Control, and Dynamics (Oct. 1 2021-Sep. 30, 2022)

SEP 2019 - AUG 2022 | Murata Overseas Foundation Scholarship

JAN 2019 - DEC 2023 | Japan Student Services Organization Scholarship

MAR 2018 | Outstanding Student Award of Japan Society for Aeronautical and Space Sciences

MAR 2015 | Kyushu University President's Award and Scholarship

TECHNICAL JOURNAL OR CONFERENCE REFEREE ACTIVITIES

- Delegate reviewer, IEEE Conference on Decision and Control, 2023
- Reviewer, AIAA Journal of Guidance, Control, and Dynamics, 2022
- Reviewer, AIAA SciTech, 2021

SELECTED COURSEWORK

Control and Robotics

Linear Control, Nonlinear Control, Optimal Control, Nonlinear Stochastic Optimal Control,
Planning and Decision-Making for Autonomy, Intelligent Cyber-Physical Systems,
Optimization-Based Learning Control and Games

Math

Real Analysis, Optimization, Statistics

Artificial Intelligence

Artificial Intelligence, Machine Learning

Astrodynamics and Space Systems

Orbital Mechanics, Space Exploration Robotics Engineering, Spacecraft Attitude Determination and Control,
Spacecraft Design, Optical Navigation (Audit)

REFERENCES

Koki Ho, Ph.D.

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Georgia Institute of Technology
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